HW #2
Musical Instrument & Pitch Detection

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Prerequisite: the following Python libraries are suggested for this assignment.

- `librosa`, a Python library for music and audio signal processing.
- `pretty-midi`, a Python library for MIDI signal processing
- `mir_eval`, a Python library for MIR evaluation
- `soundfile`, a Python library for audio I/O
- `mir-data`, an Python library that provides tools for working with MIR datasets.
- `scikit-learn`, a Python library for machine learning..

1. Instrument classification

   In this assignment, you will learn to utilize signal processing and machine learning methods to solve pitch and instrument classification problems. You also need to discuss the classification results.

   The TinySOL dataset[1] is employed for this assignment. This dataset contains 2,913 audio clips as WAV files, sampled at 44.1 kHz, with a single channel (mono), at a bit depth of 16. For education and research purposes. the dataset can be employed for training and/or evaluating instrument recognition or fundamental frequency estimation tasks. The authors of the dataset also provide an official 5-fold split of TinySOL. This split has been carefully balanced in terms of instrumentation, pitch range, and dynamics.

   (a) (20%) Below is an example of TinySol dataset's track ID annotation. Please explain the meaning of this ID.

   Track ID: Hn-ord-C4-pp-N-N.wav

   (b) (20%) Plot the spectrogram, mel-spectrogram, and CQT (using the `librosa` functions) of TpC-ord-C6-mf-N-T11u.wav and BTb-ord-F#1-mf-N-T18u.wav. Please explain the differences between these features. For spectrogram and mel-spectrogram, if you use a window size = 1024 samples, can the characteristics of both the two signal be well represented? Find the window sizes for the two audio files such that you feel suitable to represent the characteristics of them.

   (c) (30%) Use machine learning models from `scikit-learn` and implement feature extraction (i.e., spectrogram, mel-spectrogram, CQT) on the TinySol dataset to perform an instrument classification task. Use the fold partition (i.e. folds 0, 1, 2, 3, 4) provided by the TinySol dataset to perform a 5-fold cross validation.

      (a) Train a decision tree model on the three features. You may implement the model with the machine learning library `scikit-learn`, and tune the depth of the tree by to get better result.

      (b) Train a simple convolutional neural network (CNN) model on the three features. Let's consider LaNet-5, an old yet tiny model for the traditional image classification task. The model architecture of the LaNet-5 model is listed in Table 1.

Layer	Parameters
Conv2D	`filters` = 6, `kernel_size` = (5, 5)
MaxPooling2D	`pool_size` = (2, 2)
Conv2D	`filters` = 16, `kernel_size` = (5, 5)
MaxPooling2D	`pool_size` = (2, 2)
Flatten	–
Dense	`units` = 120
Dense	`units` = 84
Dense	`units` = number of classes

Table 1: Model architecture of LaNet-5

Discuss the results, including the confusion matrix, precision, recall, and F1-score. What instrument classes are difficult to classify? What instrument classes are difficult to be distinguished from other classes? Which feature representation is better?

(d) (30%) Use machine learning models from scikit-learn and implement feature extraction on the TinySol dataset to perform a pitch classification task. Discuss the results, including the confusion matrix, precision, and recall. Please implement all the three models with scikit-learn.

(a) Use the autocorrelation function (ACF).

(b) Use `librosa.pyin`, the pYIN pitch detection function in the `librosa` library.

(c) Train a simple CNN model (LaNet-5) on the three features.

Similarly to (c), compute and discuss the results, including the confusion matrix, precision, recall, and F1-score.

We suggest you refer to the start code, and apply it in this assignment: https://colab.research.google.com/drive/1Vl_DgmmN_vdE1w_4t6MbSS783OM52JR4?usp=sharing

Please submit your .zip file containing the report (.pdf) and your code (colab), with the file name "HW1_[your ID]" to the course website. Please note that, as a report, a detailed discussion on your result is encouraged. The deadline of Assignment #1 is May 6.

[1] https://zenodo.org/record/3685331